



## SEMI-LUG BUTTERFLY VALVES

### Chem-Tek Series

Corrosion Resistant  
Stainless Steel Body

Model: 53 Series  
Size Range: 2" thru 12"  
Model 53A Sizes 6"-12"  
Pressure Rating  
BUNA Seat: 200 WOG  
EPDM Seat: 200 WOG  
VITON Seat: 200 WOG  
PTFE Seat: 150 WOG



Sizes  
2" - 6"

Sizes - 8", 10" & 12" Featured  
with Actuator Direct Mount ISO  
mountings & adjustable V-Ring  
packing.

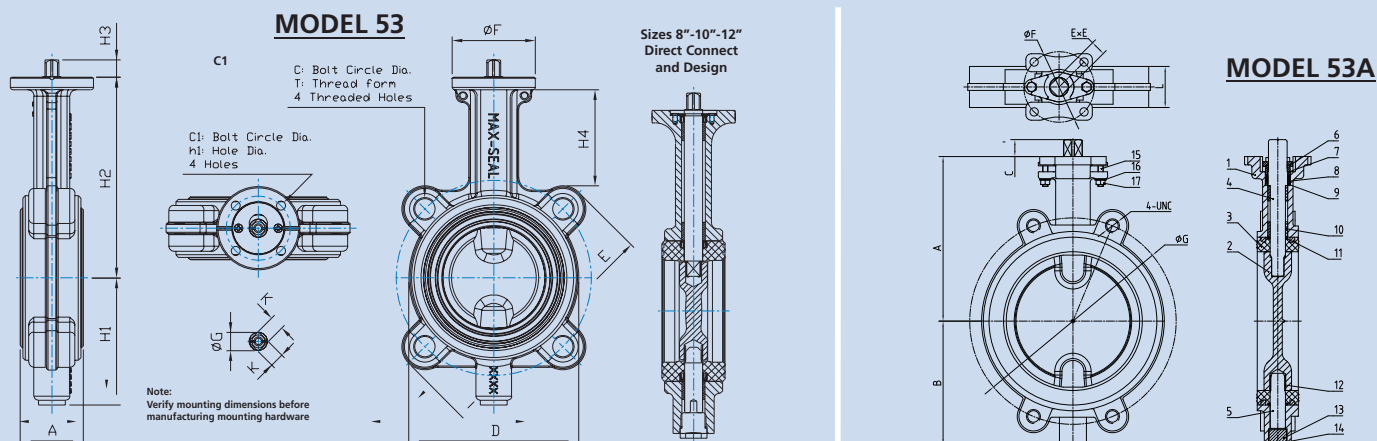
### Design Advantages

- Semi-Lug Body Design Crossover from Wafer to Lug. All Valves Have Four Drilled and Tapped Lugs
- Lower Cost than Full Lug Bodies
- Easy Assembly and Maintenance
- Ideal for Exact Alignment
- Lighter Weight than Full Lug Bodies Remain in Piping Systems when One Side of the Piping is Removed

Max-Seal Concentric Butterfly Valves are backed by the resources and experience of over thirty-five years of process valve and automation experience.

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| Class 150 Concentric Butterfly Valves |      |       |       |      |      |       |      |      |      |      | Flange Dimension |           | Mounting Base ISO 5211 |      |     | Weight |
|---------------------------------------|------|-------|-------|------|------|-------|------|------|------|------|------------------|-----------|------------------------|------|-----|--------|
| SIZE                                  | A    | D     | E     | F    | H1   | H2    | H3   | H4   | G    | K    | C                | T         | C1                     | H1   | ISO | lb     |
| 2"                                    | 1.69 | 3.98  | 6.09  | 2.56 | 2.83 | 5.01  | 0.53 | 2.22 | 0.55 | 0.43 | 4.75             | 5/8-11UNC | 1.97                   | 0.28 | F05 | 5.29   |
| 2.5"                                  | 1.81 | 4.50  | 6.84  | 2.56 | 3.23 | 5.53  | 0.53 | 2.27 | 0.55 | 0.43 | 5.50             | 5/8-11UNC | 1.97                   | 0.28 | F05 | 6.90   |
| 3"                                    | 1.81 | 5.09  | 7.42  | 2.56 | 3.76 | 6.23  | 0.53 | 2.97 | 0.55 | 0.43 | 6.00             | 5/8-11UNC | 1.97                   | 0.28 | F05 | 7.98   |
| 4"                                    | 2.05 | 6.17  | 8.84  | 3.54 | 4.47 | 6.63  | 0.69 | 2.01 | 0.71 | 0.55 | 7.50             | 5/8-11UNC | 2.76                   | 0.35 | F07 | 11.60  |
| 5"                                    | 2.20 | 7.48  | 9.92  | 3.54 | 5.08 | 7.15  | 0.69 | 2.01 | 0.71 | 0.55 | 8.50             | 3/4-10UNC | 2.76                   | 0.35 | F07 | 14.42  |
| 6"                                    | 2.20 | 8.35  | 11.23 | 3.54 | 5.59 | 8.05  | 0.73 | 2.22 | 0.87 | 0.67 | 9.50             | 3/4-10UNC | 2.76                   | 0.35 | F07 | 19.18  |
| 8"                                    | 2.36 | 10.40 | 13.60 | 4.92 | 6.77 | 9.04  | 0.96 | 2.03 | 1.11 | 0.87 | 11.75            | 3/4-10UNC | 4.02                   | 0.43 | F10 | 28.26  |
| 10"                                   | 2.68 | 12.59 | 16.10 | 4.92 | 8.39 | 10.53 | 0.96 | 2.07 | 1.11 | 0.87 | 14.25            | 7/8-9UNC  | 4.02                   | 0.43 | F10 | 39.73  |
| 12"                                   | 3.07 | 14.69 | 18.85 | 5.91 | 9.53 | 11.54 | 1.04 | 2.01 | 1/43 | 1.06 | 17.00            | 7/8-9UNC  | 4.92                   | 0.51 | F12 | 58.47  |

- Larger sizes available through 24"
- Pressure Ratings: 2"-12" 200 PSI
- Dead End Service: 2"-3" 80 psi; 4" -12" 60psi
- Vacuum Service up to 28" Hg
- Max- Seal valves are designed for bubble tight shut off either direction of flow. Each valve is factory tested 110% of their pressure rating.
- Blow out proof stem design
- A Corrosion resistant butterfly valve, designed for ANSI Class 125/150 Flanges. These valves comply with MSS- SP25, MSS-SP67 and API609 specifications, as well as meeting the requirements of MIL-V-22133C(ship)Type 1, Class A-D
- Positive Valve Position: When the handle is perpendicular to the pipe, the valve is shut. When the handle is parallel to the pipe, the valve is fully open. The orientation of the disc is indicated by a groove on the shaft end that is in line with the disc.

#### BILL OF MATERIALS

| No. | Part name               | Material       |
|-----|-------------------------|----------------|
| 1   | Body                    | ASTM A351 CF8M |
| 2   | Disc                    | ASTM A351 CF8M |
| 3   | Seat                    | EPDM           |
| 4   | Stem                    | 17-4PH         |
| 5   | Lower stem              | 17-4PH         |
| 6   | Packing gland           | ASTM A351 CF8M |
| 7   | Packing ring            | SS304          |
| 8   | Packing                 | RPTFE          |
| 9   | Packing supporting ring | SS304          |
| 10  | Bearing                 | RPTFE          |
| 11  | O ring                  | EPDM           |
| 12  | Bearing                 | RPTFE          |
| 13  | Plug                    | SS304          |
| 14  | Bottom cover seal       | EPDM           |
| 15  | Bolt                    | SS304          |
| 16  | Belleville              | SS304          |
| 17  | Nut                     | SS304          |

| DN  | INCH | A   | B   | C    | E     | F   | L  | G     | UNC       |
|-----|------|-----|-----|------|-------|-----|----|-------|-----------|
| 50  | 2"   | 126 | 78  | 13.5 | 11x11 | F05 | 43 | 120.7 | 5/8-11UNC |
| 65  | 2.5" | 134 | 84  | 13.5 | 11x11 | F05 | 46 | 139.7 | 5/8-11UNC |
| 80  | 3"   | 145 | 92  | 13.5 | 11x11 | F05 | 46 | 152.4 | 5/8-11UNC |
| 100 | 4"   | 170 | 114 | 17.5 | 14x14 | F07 | 52 | 190.5 | 5/8-11UNC |
| 125 | 5"   | 185 | 129 | 17.5 | 14x14 | F07 | 56 | 215.9 | 3/4-10UNC |
| 150 | 6"   | 203 | 144 | 18.5 | 17x17 | F07 | 56 | 241.3 | 3/4-10UNC |
| 200 | 8"   | 239 | 179 | 24.5 | 22x22 | F10 | 60 | 298.5 | 3/4-10UNC |
| 250 | 10"  | 275 | 216 | 24.5 | 22x22 | F10 | 68 | 362   | 7/8-9UNC  |
| 300 | 12"  | 307 | 247 | 30   | 27x27 | F12 | 78 | 431.8 | 7/8-9UNC  |

Max-Seal Chem-Tek Series offers longer service life, greater reliability, ease of parts replacement and interchangeability of components.

## Model 53 Semi-Lug Body

**Four Threaded Body Lugs, Threads can be removed if desired**

## Investment Cast Stainless Body

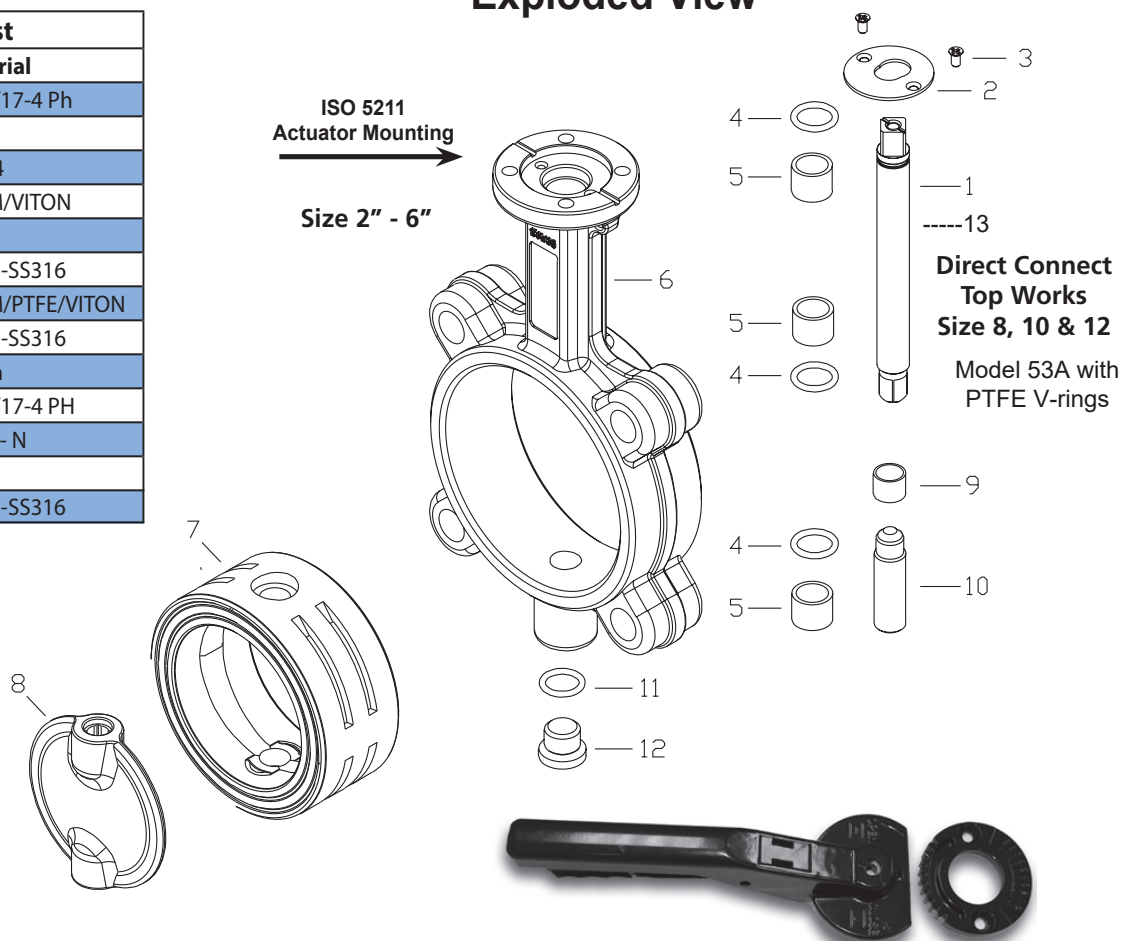
## Design Features

One piece body with extended neck allows clearance for flanges and up to 2 inches or more of insulation.

Primary stem seals are formed by preloaded contact between the disc and seat. A secondary seal is effected by having a stem diameter greater than the stem hole in resilient seat.

Field replaceable, phenolic bonded cartridge seat is blowout proof, stretch resistant and non-collapsible, making it an ideal seat design for high velocity or vacuum service. Our resilient seat design eliminates the need for flange gaskets.

## Exploded View



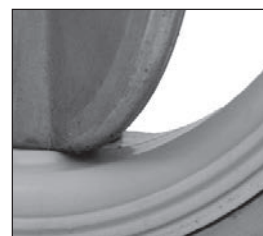
### Unique 10-position Locking Handle

- Body and Disc, all sizes, are made by investment casting.
- High strength, square drive (2" - 12") ensures a positive shaft to disc connection.
- Disc floats inside the seat for positive sealing and extended seat life.
- No pins or bolts exposed to flow.

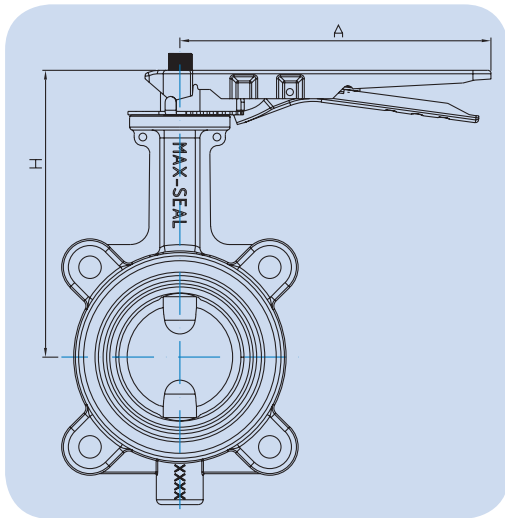
## Optimal Performance Design

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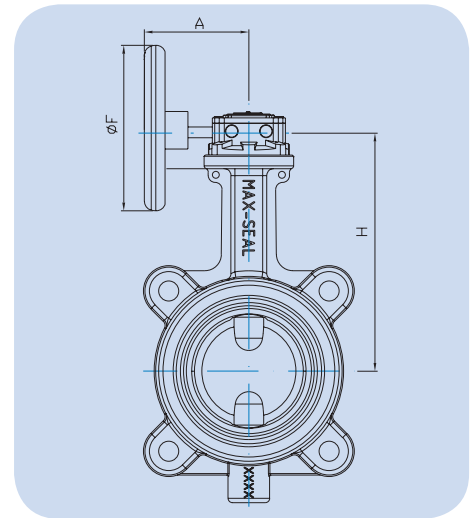
| Ordering Example by Part Number |               |               |               |               |                  |          |      |   |   |   |   |     |
|---------------------------------|---------------|---------------|---------------|---------------|------------------|----------|------|---|---|---|---|-----|
| Semi Lug                        | Stainless 316 | Stainless 316 | Stainless 316 | Buna          | EPDM             | Lever    | Size |   |   |   |   |     |
| Model                           | Body Material | Disc Material | Stem Material | Seat Material | Stem Seal O-Ring | Operator | 10"  |   |   |   |   |     |
| 53                              | -             | SS            | -             | SS            | -                | B        | -    | E | - | L | - | 250 |



**Accurate and smooth machined profile of disc edge requires minimal deformation of the resilient elastomer liner to achieve a positive seal. The low deformation results in low torque, less wear of the seat liner and increased operational life.**



All Types of  
Pneumatic &  
Electric Actuators.  
Consult Factory



| Lever-Lock 10 Position Type Handle |      |      |      |       |       |       |       |       |       |
|------------------------------------|------|------|------|-------|-------|-------|-------|-------|-------|
| Size                               | 2"   | 2.5" | 3"   | 4"    | 5"    | 6"    | 8"    | 10"   | 12"   |
| H                                  | 6.49 | 6.81 | 7.71 | 8.22  | 8.74  | 9.64  | 10.89 | 12.38 | 13.39 |
| A                                  | 8.66 | 8.66 | 8.66 | 10.24 | 10.24 | 10.24 | 14.57 | 14.57 | -     |

Standard handles through 6" are aluminum. Other handle materials are available. Please consult factory.

| Gear Operator |      |      |      |      |      |      |       |       |       |
|---------------|------|------|------|------|------|------|-------|-------|-------|
| Size          | 2"   | 2.5" | 3"   | 4"   | 5"   | 6"   | 8"    | 10"   | 12"   |
| H             | 5.88 | 6.20 | 7.10 | 7.73 | 8.25 | 9.15 | 10.34 | 11.83 | 12.84 |
| F             | 3.94 | 3.94 | 4.72 | 4.72 | 4.72 | 7.87 | 11.81 | 11.81 | 11.81 |
| A             | 4.25 | 4.25 | 4.25 | 4.69 | 4.69 | 4.69 | 8.78  | 8.78  | 8.78  |

## VALVE FLOW COEFFICIENTS & TECHNICAL DATA

| Seat Temperature Range |                |
|------------------------|----------------|
| Liner                  | Temperature    |
| EPDM-H                 | 5 to 250 °F    |
| NBR-Buna-N             | 14 to 200 °F   |
| Natural Rubber         | - 22 to 176 °F |
| Hypalon                | 0 to 167 °F    |
| Silicone               | - 40 to 338 °F |
| Viton                  | 0 to 380 °F    |
| PTFE Silicone Backed   | - 40 to 356 °F |

| Chem-Tek Series Torque Valve |     |                       |         |         |
|------------------------------|-----|-----------------------|---------|---------|
| Size                         |     | Pressure Differential |         |         |
| Inch                         | MM  | 50 psi                | 100 psi | 200 psi |
| 2"                           | 50  | 79                    | 92      | 106     |
| 2.5"                         | 65  | 111                   | 139     | 159     |
| 3"                           | 80  | 169                   | 185     | 212     |
| 4"                           | 100 | 258                   | 285     | 327     |
| 5"                           | 125 | 424                   | 489     | 522     |
| 6"                           | 150 | 637                   | 769     | 885     |
| 8"                           | 200 | 1234                  | 1539    | 1770    |
| 10"                          | 250 | 1661                  | 2155    | 2478    |
| 12"                          | 300 | 2251                  | 2971    | 3416    |

Consult Factory for  
Teflon Seated Torques

Optional Seat Materials are also  
Available C/F

| Max Seal Models 53 CV Value |     |                  |     |     |     |      |      |      |      |      |
|-----------------------------|-----|------------------|-----|-----|-----|------|------|------|------|------|
| Size                        |     | Angle of Opening |     |     |     |      |      |      |      |      |
| Inch                        | MM  | 10°              | 20° | 30° | 40° | 50°  | 60°  | 70°  | 80°  | 90°  |
| 2"                          | 50  | 1.1              | 1.1 | 5.3 | 15  | 30   | 49   | 75   | 103  | 112  |
| 2.5"                        | 65  | 1.1              | 2.1 | 12  | 28  | 53   | 81   | 128  | 180  | 224  |
| 3"                          | 80  | 3.2              | 6.3 | 29  | 57  | 96   | 147  | 224  | 316  | 424  |
| 4"                          | 100 | 5.3              | 15  | 60  | 113 | 184  | 275  | 424  | 624  | 839  |
| 5"                          | 125 | 6.3              | 28  | 88  | 164 | 260  | 404  | 665  | 1002 | 1301 |
| 6"                          | 150 | 7.4              | 54  | 135 | 235 | 381  | 601  | 1026 | 1612 | 2026 |
| 8"                          | 200 | 23               | 120 | 241 | 421 | 671  | 1069 | 1843 | 3025 | 3659 |
| 10"                         | 250 | 35               | 180 | 351 | 666 | 1019 | 1607 | 2783 | 4624 | 6042 |
| 12"                         | 300 | 51               | 263 | 515 | 972 | 1487 | 2343 | 4059 | 6975 | 9272 |

Streamlined Disc Design Reduces Pressure Drop and Maximizes Cv

## Actuator Torques

- The torque ratings in this brochure should be considered as a guide and not as an unqualified recommendation.
- Torque figures are for wet lubricated with clean non-abrasive line media.
- Temperature well within resilient seat limits
- Valve to be operated a minimum of once a day
- For applications which require clarification or additional information, consult factory.

Max-Seal Valves Are Ideally Suited for  
Actuated Applications

*Direct Mount Sizes 8", 10", 12"*



## ACTUATOR MOUNTING

Max-Seal offers a broad line of automation systems for precise proportioning or on-off control in either pneumatic or electrically powered units. Cast mounting flange accommodates all types of operators, offering ISO5211 mounting bolt circle. It is designed to accept direct actuator mounting, some sizes may require a spacer plate. Direct mounting eliminates exposed linkages, bracket and adapter misalignment and bolt corrosion. It also provides a stronger, more compact valve package.



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